ReSkip: Retrofitting Intrinsic Depth Routing into Pretrained Transformers via Attention Residuals

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Abstract

Adaptive computation is a central principle of biological intelligence: simple stimuli can often be processed by shallow feedforward pathways, whereas difficult stimuli recruit deeper and recurrent computation. Existing depth-adaptive transformers explore this idea through early exits, token routers, or modified pretraining, but post-hoc retrofitting of frozen pretrained decoders into adaptive-depth models remains underexplored. We propose AR-RETROFIT, a lightweight method that installs an intrinsic depth-routing signal into an existing decoder. Starting from an exact identity at initialization, AR-RETROFIT adds a γ-gated ATTNRES-routed correction between frozen backbone blocks, so the original full-depth path is preserved while the routed path becomes active during a short fine-tune. The retrofit uses $\sim 0.7\%$ new parameters on Qwen3-VL-2B. Because the routing weights are computed before each block executes, they can also drive RESKIP, a calibrated rule for input-dependent block skipping. In a 340M controlled model, RESKIP matches full-depth benchmark scores while giving about 20% lossless wall-clock speedup over full-depth ATTNRES. On Qwen3-VL-2B and Qwen3-VL-4B, AR-RETROFIT improves $5/6$ VLM benchmarks at both scales, with mean gains of $+2.1$pp and $+1.2$pp, raises LAMBADA by $+3.3$pp and $+8.7$pp, and keeps compiled full-depth runtime within 2.9% of the base. When used to initialize LIBERO vision-language-action policies, the retrofitted backbones improve matched OpenVLA-OFT baselines by $+1.30$pp and $+0.65$pp, while conservative RESKIP preserves action-stream success rates. These results show that existing pretrained decoders can acquire practical depth-axis adaptive computation through a low-cost retrofit, without training an adaptive architecture from scratch.

1 Introduction

Different inputs should not always spend the same computation. This is a design motivation rather than a claim of cognitive equivalence. Kahneman's dual-process framework [19] distinguishes fast and deliberate processing, and biological vision offers a concrete computational analogue: easy stimuli can be recognised through shallow feedforward sweeps, while harder stimuli recruit recurrent and deeper circuits [26, 20, 10]. Recurrent vision models capture human representational dynamics [22], and confidence-thresholded recurrence spends more steps on harder images to trade speed for accuracy [42]. At a more local scale, cortical association and dendritic gating provide mechanistic examples of computation-dependent routing inside the processing substrate rather than in a separate controller [3, 6, 27]. Since finite recurrence can be unrolled into feedforward depth [44], these observations motivate a transformer-side question: can a pretrained model acquire an intrinsic signal for allocating effective depth per input?

Large language models already expose one test-time compute axis: *tokens*. Test-time compute scaling [34, 8, 41], chain-of-thought prompting [45], and self-thought methods [48] spend more compute on hard problems by emitting more reasoning tokens before answering. Yet every token in such a trace still traverses every layer of the underlying transformer, regardless of whether that token is a routine connector or a load-bearing inferential step. The missing complementary axis is therefore *depth*: deciding which blocks to invoke for this input, at this point in the sequence.

Existing depth-axis methods do not quite match this motivation because they externalise the decision. Early-exit classifiers [40, 12] add auxiliary heads per layer; Mixture-of-Depths [39] learns a token router with a capacity-balancing loss; layer-pruning [16, 32] is post-hoc and static; Universal Transformers [9] share weights with ACT-style halting [15]. In each case a separate component decides whether the rest of the network should run. The biological analogue motivating this paper is different: control is embedded in the processing substrate rather than attached as a separate decision head.

ATTNRES [25] gives a natural handle. It replaces the fixed scalar-1 residual with a learned softmax-attention over previous block outputs; the routing weights $\alpha_{i\to n}$ are input-dependent, competitively normalised, and computed *before* block n runs, as part of forming its input. They therefore emit a per-block depth-allocation signal as a side-effect of the network's own forward, not as a separate head. The practical obstacle is that existing ATTNRES models are trained from scratch with the modified residual; reproducing this at the 2–7B scale that real applications need costs tens of thousands of GPU-hours (Appendix D.1). The central problem in this paper is how to install such a signal into standard pretrained transformers through a short fine-tune, while preserving their original capabilities and making the signal usable for calibrated depth skipping.

The paper has one main contribution and two supporting claims.

- **AR-RETROFIT installs intrinsic depth routing into a pretrained model** (§2.2). A γ-gated residual injection adds an ATTNRES-style routing path to a frozen transformer through a short fine-tune, using $\sim$0.7% new parameters and an identity-preserving initialization.
- **RESKIP uses the installed signal for pre-execution dynamic depth** (§2.3, §3, §4.2). The skip decision is made before the current block runs; a 340M controlled study verifies the speed-quality motivation, and eligible blocks are selected by ablation or action-drift because routing weight is a useful signal but not a safety certificate.
- **VLM and VLA experiments validate the retrofit pathway** (§4, §5). The retrofit preserves or improves pretrained VLM capability at near-base full-depth runtime, dynamic RESKIP gives calibrated adaptive-depth behavior across q settings, and LIBERO provides supporting transfer evidence.

2 Method

2.1 Attention Residuals as a pre-execution routing signal

A standard transformer uses fixed scalar-1 combination, $\mathbf{x}_l = \mathbf{x}_{l-1} + f_l(\mathbf{x}_{l-1})$, emitting no routing signal. ATTNRES [25] replaces this with a learned attention over depth. Each block l maintains a pseudo-query $\mathbf{w}_l \in \mathbb{R}^d$; keys are projections $\mathbf{k}_i = W_K \mathbf{h}_i$ of earlier block outputs $\mathbf{h}_i$, and

$$\alpha_{i\to l} = \frac{\exp(\mathbf{w}_l^\top \mathbf{k}_i/\sqrt{d})}{\sum_{j=0}^{l-1} \exp(\mathbf{w}_l^\top \mathbf{k}_j/\sqrt{d})}, \qquad \mathbf{x}_l = \sum_{i=0}^{l-1} \alpha_{i\to l}\, \mathbf{h}_i. \tag{1}$$

The key property is two-phase execution. Computing the input to block l requires only $\{\mathbf{h}_i\}_{i<l}$ and $\mathbf{w}_l$—both available *before* f_l runs. We call the pre-execution attention computation *phase 1* and f_l itself *phase 2*. Phase 1 is the routing signal we exploit. Block-ATTNRES groups layers into N blocks and applies the softmax at the block level, which is the granularity used throughout. Algorithm 1 gives the full two-phase forward with RESKIP's skip check folded in.

2.2 AR-RETROFIT: installing the routing path

The retrofit target is constrained by the problem in the introduction: take a transformer pretrained in the standard-residual regime and produce an ATTNRES-capable model that (P1) preserves benchmark quality, (P2) emits a routing α informative enough to drive RESKIP, and (P3) trains in a single short fine-tune over off-the-shelf SFT data, with the

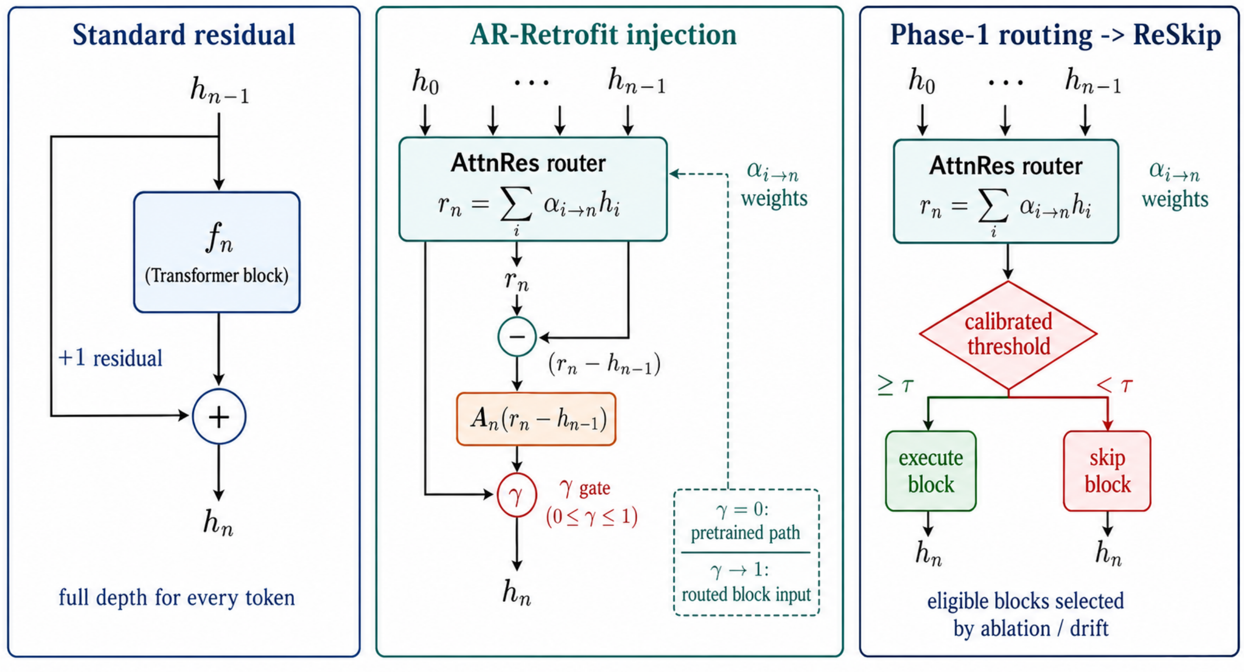

Figure 1: **Overview of ReSkip and AR-Retrofit.** Left: a standard transformer residual gives no intrinsic routing signal. Middle: AR-Retrofit injects an identity-preserving γ-gated AttnRes path into the frozen pretrained backbone. Right: the phase-1 AttnRes weights are computed before block execution and are calibrated into ReSkip decisions.

base frozen. To satisfy the first constraint, the modification must be exactly identity-preserving at step 0; to satisfy the second, the routing must feed the real forward path, not merely observe it.

We organise the decoder into N *blocks* of adjacent layers (Qwen3-VL-2B uses 7 blocks of 4 layers in the main experiments). For every block $n \geq 1$ we add an AttnRes router $\mathbf{w}_n$, a tiny adapter A_n (down-up bottleneck of rank r, SiLU), and a scalar gate γ_n:

$$r_n = \sum_{i=0}^{n-1} \alpha_{i\to n}\, h_i, \qquad x_n = h_{n-1} + \gamma_n\, A_n(r_n - h_{n-1}), \qquad h_n = \mathrm{Block}_n(x_n). \tag{2}$$

Block_n is the pretrained block group, verbatim; AttnRes enters *between* blocks, as a correction. The retrofit parameters $\{\mathbf{w}_n, \gamma_n, A_n\}$ amount to $\sim$15M ($\sim$0.7% of base) at r=256 on 2B; the entire base—embeddings, vision tower, decoder layers, LM head—is frozen.

With γ_n=0 we have $x_n = h_{n-1}$ exactly, so the forward at step 0 is bit-identical to the pretrained model. The adapter up-projection is small-random ($\mathcal{N}(0, 0.02^2)$) so that $\partial\mathcal{L}/\partial\gamma_n \neq 0$ at init, avoiding gradient deadlock. We then ramp γ_n via a linear curriculum $0 \to 1$ over the first 30–50% of training. At γ_n=1, the AttnRes-routed correction is fully active, but the model remains a residual correction on the pretrained stream as defined in Eq. 2. Appendix C.1 compares observer-only, interpolation, and informed-initialisation controls.

The training objective follows from the three constraints above. We minimise

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(\text{full path}) + \lambda_{\mathrm{kl}}\, \mathcal{L}_{\mathrm{KL}}(\text{skip path} \,\|\, \text{teacher}) + \lambda_{\mathrm{ent}}\, \mathcal{L}_{\mathrm{ent}}(\alpha), \tag{3}$$

where the full-path CE is computed on assistant tokens, while the skip-branch KL samples one block at random per step, runs the forward with it skipped, and pulls the resulting logits toward a frozen pretrained teacher. This makes the surrogate x_n useful both when the block runs and when it is skipped. $\mathcal{L}_{\mathrm{ent}}$ is implemented with a negative sign in the loss, i.e., a light entropy-maximising prior against premature collapse of α. Default λ_{kl}=1, entropy weight 0.02, lr 10^{-3} on retrofit params, cosine schedule with 100-step warmup. Hyperparameters and data mixtures are listed in Appendix B.3.

This is not ordinary SFT. The CE term keeps the full path useful for the downstream task, but by itself it does not train the skipped surrogate to replace a real block. The skip-branch KL supplies that constraint by tying one skipped-block forward per step to the frozen teacher. Conversely, teacher imitation alone would merely reproduce the base model. The weak entropy prior prevents early one-source collapse while still allowing the router to sharpen later. This three-term structure is what makes the retrofit identity-preserving at step 0, task-improving after training, and skip-ready at inference.

2.3 ReSkip: dynamic block skipping from the installed signal

ReSkip uses the installed routing path rather than a separate controller. For block n, phase 1 computes $\alpha_{\cdot\to n}$ before the block body runs. We then read the weight $w_{\text{recent}}(n)$ that the router assigns to the immediate predecessor and decide whether to execute phase 2:

$$\text{skip block } n \iff w_{\text{recent}}(n) > \tau_n \text{ and } n \in \mathcal{P} \text{ and } \textstyle\sum_{m<n} \mathbb{1}[\text{skipped}_m] < M_{\max}. \tag{4}$$

The decision is available before the current block executes, so a trigger can actually save that block. When the rule fires, we short-circuit the block as $h_n \leftarrow x_n$, where x_n is the routed-correction input in Eq. 2. To keep autoregressive decoding cache-consistent, a skipped decoder layer still runs the K/V-producing slice (`LayerNorm`$\to$`k/v_proj`$\to$`RoPE`$\to$`cache.update`) and skips `q_proj`, attention, `o_proj`, and MLP. The cache-enabled skip path matches the cacheless path in last-position argmax (Appendix B.6).

2.4 Calibration and safety selection

Routing weight is a usable depth signal, not a certificate that a block can be removed. Each eligible set $\mathcal{P}$ is therefore selected offline from two complementary diagnostics. The AttnRes importance score

$$I(n) = \max_{l>n} \mathbb{E}[\alpha_{n\to l}]$$

measures how often future blocks refer to block n, while an ablation or drift score measures how costly it is to remove that block on the relevant distribution. For language experiments we use $A(n) = \text{PPL}(n \text{ removed})/\text{PPL}(\text{full})$; for VLA we use action-drift MSE on simulated rollouts. Thresholds τ_n are the q-th empirical quantile of $w_{\text{recent}}(n)$ on held-out calibration data. This separation is central: α decides *when* a calibrated block looks locally redundant, while ablation or drift decides *which* blocks are eligible to skip.

3 Controlled Motivation: AttnRes and ReSkip at 340M

3.1 AttnRes and ReSkip at 340M

We first test the mechanism in the setting where AttnRes is trained from scratch, before introducing any retrofit confound. On a 340M block-AttnRes transformer (d=1024, L=24, N=8 blocks; FineWeb-Edu 100BT [36]; `lm-eval-harness` [14]), ReSkip at $\mathcal{P}$={3, 5}, $M_{\max}$=2, q=0.85 matches the full-depth AttnRes scores across the expanded language benchmark set and gives 1.19× wall-clock at sequence length 8192 (Tab. 1; Pareto curves in Appendix D.6). The observed skip count averages 0.88 blocks and varies 0–2 per forward, confirming input-dependent allocation rather than a threshold that never fires.

This controlled study establishes the motivation for the retrofit experiments: AttnRes improves the vanilla residual baseline, exposes a pre-execution depth signal, and lets ReSkip recover wall-clock speed without changing the trained weights. We return to matched-rate and calibration ablations in Section 6.

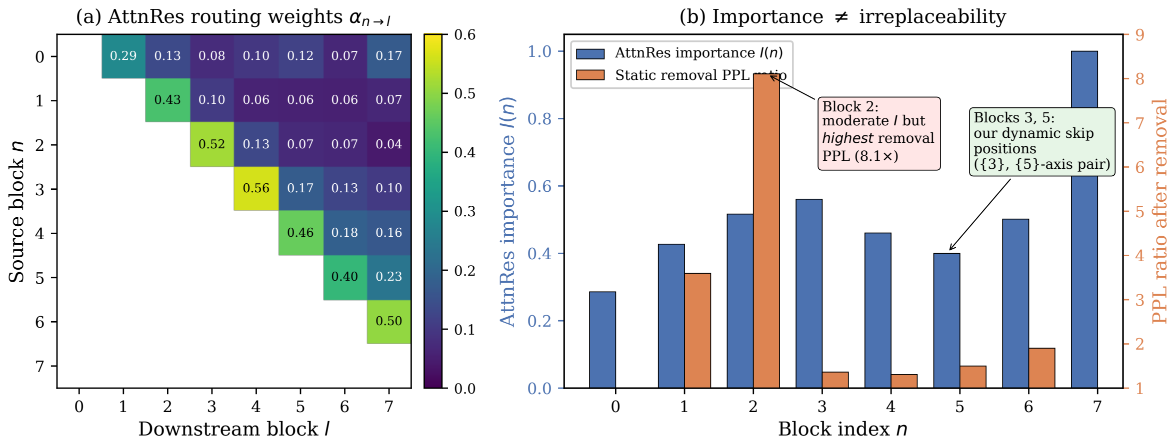

Figure 2: **ATTNRES routing weights do not predict ablation impact (340M).** (a) Learned block-level $\alpha_{n\to l}$. (b) $I(n)$ (blue, left axis) and $A(n)$ (orange, right axis). RESKIP uses routing for input-dependent decisions, but eligible blocks still need ablation-based safety selection.

Table 1: **RESKIP on 340M from-scratch ATTNRES.** The selected operating point preserves the full-depth ATTNRES scores on the expanded task set while actually firing: ReSkip averages 0.88 skipped blocks over 48 sequence-8192 batches and gives 1.19× wall-clock at that length. Headline: `acc_norm` for PIQA/HellaSwag/ARC-E/ARC-C/OpenBookQA, plain `acc` for MMLU/LAMBADA.

Configuration	Avg. skip	Speed	LAMBADA acc	LAMBADA ppl	HellaSwag	PIQA	ARC-E	ARC-C	MMLU	OpenBookQA
Vanilla (no ATTNRES)	–	–	0.3790	24.71	0.4436	0.6779	**0.5602**	**0.3046**	**0.2594**	0.3320
ATTNRES full-depth	0.00	1.00×	0.4054	20.20	0.4607	0.6893	0.5438	0.3012	0.2555	0.3580
+ RESKIP ($\{3, 5\}$, M=2, q=0.85)	**0.88**	**1.19×**	**0.4054**	**20.20**	**0.4607**	**0.6893**	**0.5438**	**0.3012**	**0.2555**	**0.3580**

4 Retrofitting Qwen3-VL: VLM Quality, Runtime, and ReSkip

4.1 VLM adaptation: quality against the base

We retrofit **Qwen3-VL-2B** [2] (L=28, d=2048) at N=7 blocks of L=4 and **Qwen3-VL-4B** (L=36, d=2560) at N=9 blocks of L=4. All base parameters are frozen; ∼15M (2B) / ∼23.7M (4B) retrofit parameters at r=256 train via Eq. 3. The final SFT mixture contains 60% LLaVA-OneVision-Data [28], 20% UltraChat [11], 10% NuminaMath-CoT [1], and 10% OpenThoughts-114k [17] for 10k steps; Appendix C.6 gives the concrete mixture. Evaluation uses LAMBADA [35]/HellaSwag [49] for text and six `lmms-eval` benchmarks for VLM (MMBench [30], MMMU [47], MMStar [7], AI2D [21], OCRBench [31], RealWorldQA [46]). Identity-at-init is verified before training (γ=0 reproduces base; max $|\Delta\text{logits}| = 0.375$ bf16, 100% argmax agreement; Appendix B.5).

The retrofit improves base on five of six VLM benchmarks at both scales, averaging +2.1pp across the six VLM benchmarks on 2B and +1.2pp on 4B. The largest gain appears on the MMStar math subcategory (+7.9pp at 2B, +3.9pp at 4B; full 6-cell decomposition in Appendix E.7). The win concentrates on *deliberate reasoning over images* rather than perception, consistent with a router that lifts deeper reasoning paths. Text quality also lifts: +3.3pp LAMBADA / −16% ppl on 2B and +8.7pp / −32% on 4B; HellaSwag is within ±1.1pp at both scales. Auxiliary parameter-matched LoRA controls on a related SFT mix do not recover the LAMBADA gain (Appendix E.1). These controls do not prove that all quality gains come uniquely from routing, but they rule out a simple parameter-count-only explanation. We therefore treat the routed path as load-bearing for skip-ready behavior, while interpreting quality gains as consistent with, but not uniquely attributable to, the routed correction.

Table 2: **AR-RETROFIT on Qwen3-VL-2B/4B at the canonical L=4 recipe** (`lmms-eval` full splits for VLM, n=2000 for LAMBADA/HellaSwag). The retrofit improves base on $5/6$ VLM benchmarks at both scales, with mean VLM gains of $+2.1$pp on 2B and $+1.2$pp on 4B, and lifts text on LAMBADA. MMStar overall holds; the math subcategory (Appendix E.7) jumps $+7.9$pp on 2B / $+3.9$pp on 4B, consistent with a router that lifts deliberate-reasoning paths. Auxiliary parameter-matched LoRA controls do not show the same text-gain pattern (Appendix E.1).

	Text		VLM (`lmms-eval` full split)					
Configuration	LAMBADA	HellaSwag	MMBench	MMMU	MMStar	AI2D	OCRBench	RWQA
Qwen3-VL-2B (base)	0.532	0.506	75.77	0.414	0.536	0.736	0.772	0.648
+ AR-RETROFIT (L=4, 10k)	**0.5650**	0.500	**78.87**	**0.432**	0.536	**0.758**	**0.814**	**0.661**
Δ vs. base	**+3.3**	-0.6	**+3.10**	**+1.8**	0.0	**+2.2**	**+4.2**	**+1.3**
Qwen3-VL-4B (base)	0.576	0.562	83.33	0.490	0.624	0.819	0.819	0.715
+ AR-RETROFIT (L=4, 10k)	**0.6625**	0.5515	**85.22**	**0.521**	**0.632**	**0.825**	**0.824**	**0.718**
Δ vs. base	**+8.7**	-1.1	**+1.9**	**+3.1**	**+0.8**	**+0.6**	**+0.5**	**+0.3**

Table 3: **Quality–skip behavior after retrofit.** LAMBADA-500 on the canonical L=4 2B retrofit, $\mathcal{P}$=$\{1, 4\}$, $M_{\max}$=2, with thresholds calibrated on 32 held-out LAMBADA prefixes. The conservative point preserves quality while producing input-dependent block skipping; lower thresholds show the expected calibration boundary.

Configuration	LAMBADA acc	LAMBADA ppl	Δ acc vs no-skip	Avg. skipped blocks
No-skip retrofit	0.5700	4.526	—	0.00
AR-RETROFIT + RESKIP (q=0.85)	0.5600	5.258	-1.0pp	0.19
AR-RETROFIT + RESKIP (q=0.50)	0.4120	12.550	-15.8pp	1.06
AR-RETROFIT + RESKIP (q=0.30)	0.3900	14.005	-18.0pp	1.17

4.2 Runtime and calibrated ReSkip after retrofit

After AR-RETROFIT installs the routing path, RESKIP can be applied without adding a separate router. On the canonical 2B retrofit, LAMBADA-500 at the conservative language operating point stays within 1.0pp of the no-skip retrofit while skipping input-dependent blocks (Tab. 3). The aggressive thresholds show the expected calibration boundary: once the threshold instructs the model to skip blocks it normally executes, quality drops sharply. Together with the runtime characterization in Tab. 4, this shows the target-scale behavior we want: the installed routing path preserves or improves model capability at near-base full-depth runtime, and RESKIP turns that path into calibrated input-dependent depth allocation.

The full-depth retrofit path remains close to the base when both models run under the same fixed-shape inference setting (Tab. 4). This is the practical runtime point: adding the routing path does not erase the quality gains with a large inference penalty, and the dynamic skip rule operates on top of that installed path.

The block partition and mixture choices are reported in the appendix rather than treated as the main story. L=4 sits on the accuracy plateau alongside nearby block sizes and reduces router calls; the final mixture is the one that preserves VLM capability while training the routed correction. Full block-partition, data-mixture, and design-control ablations are in Appendices E.6, C.6, C.3.

5 VLA Transfer Evidence

A VLA test asks whether the retrofitted routed-depth signal remains usable beyond VLM evaluation. Vision tokens (perceptual, early-layer [38]), language tokens (task spec.), and action tokens (motor planning) have no principled reason to share effective depth; ATTNRES routing gives a per-position handle on this without token-level gating machinery.

We attach the OpenVLA-OFT [24] action head (L_1 regression, no chunking) to Qwen3-VL-2B/4B backbones (ViT frozen) and train on the pooled `libero_all` mix from LIBERO [29]. The main comparison is deliberately narrow: **Path 0** uses the stock backbone with OFT, while **Path B** warm-starts from our canonical L=4 VLM retrofit and keeps

Table 4: **Full-depth runtime of the installed routing path.** Forward-pass latency on $1\times$H100, bf16, batch 1, median of 20 runs after 5 warmup. The canonical compiled setting runs the retrofitted model within 2.9% of the compiled base, while improving the quality metrics in Tab. 2.

Mode	seq	base (ms)	retrofit (L=4, ms)	retrofit / base	vs uncompiled base
Uncompiled	1024	14.91	16.66	1.117×	1.117×
Uncompiled	2048	25.49	28.53	1.119×	1.119×
Compiled (`reduce-overhead`)	2048	21.31	22.50	1.056×	0.855×
Compiled (`max-autotune`)	**2048**	**21.81**	**22.43**	**1.029×**	**0.864×**

Table 5: **LIBERO 4-suite success rates (%)** for Qwen3-VL-2B/4B. Entries are success rates over 500 rollouts per suite (2,000 per policy). *Path 0* is the stock-backbone OFT baseline; *Path B* warm-starts from our L=4 VLM retrofit. The table is supporting transfer evidence for the retrofit pathway rather than a broad robotics claim.

Scale	Path (steps)	Spatial	Object	Goal	Long-10	**Avg**	Δ vs. Path 0
2B	Path 0 (30k)	94.8	99.8	97.4	91.4	95.85	—
2B	**Path B (30k, ours)**	**97.8**	99.6	**98.6**	**92.6**	**97.15**	**+1.30**
4B	Path 0 (30k)	95.0	99.2	97.8	92.2	96.05	—
4B	**Path B (30k, ours)**	94.6	**99.8**	**98.2**	**94.2**	**96.70**	**+0.65**

γ_n=1 throughout. Both paths share the same action head, data, and 30k-step OFT schedule on $4\times$H100 ZeRO-2 with effective batch 32. Each policy is evaluated for 500 rollouts per suite (2,000 per policy). The VLA in-backbone forward uses the same installed routing path as the VLM retrofit; at sequence length 2048 its eager latency is 28.79 ms vs. 25.49 ms for the stock backbone, within 1% of the VLM retrofit runtime (Appendix E.4).

5.1 LIBERO results

Path B reaches 97.15% on 2B and 96.70% on 4B. The gain concentrates on Spatial (+3.0pp at 2B) and Long-10 (+2.0pp at 4B), where we observe the clearest gains in this evaluation. We use these results as downstream transfer evidence for AR-RETROFIT, not as a broad robotics claim. Appendix F.1 reports repeated-rollout sensitivity.

5.2 RESKIP on the action stream: calibrated conservative skipping

We now apply the same RESKIP rule at inference on the trained Path B policies, with thresholds calibrated on a sim-rollout distribution rather than language tokens (the rationale is in the next paragraph). Eligible blocks are picked per scale from a per-block action-drift sweep (Appendix F.3: $\mathcal{P}=\{1,4\}$ on 2B, $\mathcal{P}=\{1,2\}$ on 4B) and $M_{\max}$=2 throughout. Tab. 6 shows the 4-suite Pareto across q.

Thresholds are calibrated on the action distribution rather than transferred from language. The reason is that q is not itself a skip rate; it indexes an empirical distribution of w_{recent} that shifts across modalities. The recommended VLA point is therefore conservative (q=0.99): it skips rarely, but only on blocks whose action-drift sweep says they are locally safe.

6 Ablations: Skip Budget and Calibration

The main knobs in RESKIP are the eligible set $\mathcal{P}$, the skip budget $M_{\max}$, and the calibration quantile q. Table 7 summarizes the operating points used in the main experiments. The 340M result shows that increasing the budget from no-skip to $M_{\max}$=2 can produce a real speedup when the skip signal is native to the trained ATTNRES model. On the retrofitted VLM and VLA, the conservative $M_{\max}$=2 settings preserve quality while using the installed routing path characterised in Tables 4 and 6.

The routing signal matters beyond the choice of skip locations. At a matched skip budget on the same 340M model, an input-dependent rule based on phase-1 routing preserves substantially more quality than static or random schedules

Table 6: **ReSkip 4-suite Pareto** on the trained Path B policies. Eligible blocks $\mathcal{P}$ and threshold τ are calibrated per scale on the action distribution; q is the empirical quantile that produces τ. The conservative end (q=0.99) preserves success rates in this evaluation.

q (sim-calibrated)	Spatial	Object	Goal	Long-10	**Avg**
Qwen3-VL-2B Path B, $\mathcal{P}=\{1,4\}$, M=2, *no-skip ref* 96.25					
0.30	4.7	—	—	—	(sharp drop)
0.85	80.0	98.0	87.3*	67.2	83.1
0.95	95.0	99.4	97.6	86.8	94.7
0.99	**97.6**	**99.2**	**99.0**	**93.6**	**97.35**
no-skip (ref)	97.4	98.6	98.0	91.0	96.25
Qwen3-VL-4B Path B, $\mathcal{P}=\{1,2\}$, M=2, *no-skip ref* 96.25					
0.30	89.6	95.4	96.4	86.0	91.85
0.50	91.2	98.0	98.2	89.6	94.25
0.85	93.6	99.2	97.6	93.2	95.90
0.95	95.6	98.0	98.0	93.0	96.15
0.99	**96.4**	**98.2**	**98.4**	**92.8**	**96.45**
no-skip (ref)	97.4	98.2	98.0	91.4	96.25

*Partial eval (332/500 trials) — driver schedule cut short; reported as the rate over completed trials.

Table 7: **Skip-budget and calibration operating points.** Quality is reported in the native metric for each setting. Runtime is measured directly for the 340M controlled model and for full-depth Qwen3-VL retrofit; the retrofitted ReSkip rows report the corresponding average skipped blocks or success-rate Pareto point.

Setting	Configuration	$M_{\max}$	q	Quality	Depth / runtime behavior
340M AttnRes	full-depth	0	–	LAMBADA 0.4054 / ppl 20.20	1.00×, 0.00 skips
340M AttnRes	ReSkip	2	0.85	LAMBADA 0.4054 / ppl 20.20	1.19×, 0.88 avg skips
2B VLM retrofit	full-depth	0	–	LAMBADA 0.5700 / ppl 4.526	22.43 ms compiled
2B VLM retrofit	ReSkip	2	0.85	LAMBADA 0.5600 / ppl 5.258	0.19 avg skips
2B VLA Path B	ReSkip	2	0.99	LIBERO avg 97.35%	conservative action-stream skipping
4B VLA Path B	ReSkip	2	0.99	LIBERO avg 96.45%	conservative action-stream skipping

(Tab. 8). The dynamic rule is not always lossless at this more aggressive calibration point, but it is clearly better than skipping the same positions unconditionally.

7 Related Work

For test-time compute, o1 [34], DeepSeek-R1 [8], chain-of-thought [45], and optimal test-time compute analyses [41] allocate compute by emitting more reasoning *tokens*; per-token depth remains fixed. Depth-axis methods include ACT / Universal Transformers [15, 9], CALM [40], Mixture-of-Depths [39], LayerSkip [13], and static pruning [16, 32]. These add a halting head, exit classifier, router, speculative decoder, or static removal rule. We instead use AttnRes as an intrinsic routing signal produced by the forward pass itself, then retrofit that signal into an existing pretrained backbone. Appendix D.5 compares dynamic, static, and random skip rules at 340M, and Appendix E.2 gives a full-layer static-pruning baseline at the VLM scale.

Table 9 summarises the closest method-level distinction. The relevant distinction is the combination of frozen-backbone retrofitting and a pre-execution signal that is also part of the forward path.

The neural motivation is recurrence as flexible effective depth. The dual-process distinction [19] maps naturally onto visual processing: easy stimuli can be handled by feedforward sweeps, while harder stimuli recruit recurrent circuits [26, 20, 10]. Recurrent networks capture human representational dynamics [22]; Spoerer et al. [42] are the

Table 8: **Dynamic, static, and random skip rules at matched rate on 340M.** All rows use the same trained AttnRes weights, $\mathcal{P}=\{3,5\}$, and $M_{\max}=1$. Dynamic ReSkip uses a calibrated phase-1 statistic; static and random controls remove the same positions without input dependence. Full decision-rule ablation and observed trigger rates are in Appendix D.5.

Rule	LAMBADA	HellaSwag	PIQA	ARC-E	OpenBookQA
Full-depth AttnRes	0.4054	0.4607	0.6893	0.5438	0.3580
Dynamic ReSkip	**0.3445**	**0.4471**	**0.6774**	**0.5391**	**0.3540**
Static skip $P=3$	0.2624	0.3968	0.6610	0.5206	0.3300
Static skip $P=5$	0.2189	0.4223	0.6638	0.5253	0.3260
Random $\{P=3$ or $P=5\}$	0.2416	0.4028	0.6420	0.5101	0.3300

Table 9: **Closest method comparison.** The relevant distinction is the combination of frozen-backbone retrofitting and a pre-execution signal that is also part of the forward path.

Method	Pretrained backbone	Extra decision head/router	Decision before current block	Applies to frozen pretrained backbone	Same signal forms input and skip
Early exit / CALM	yes	yes	no	no	no
Mixture-of-Depths	train-time design	yes	yes	no	no
Static pruning	yes	no	yes	yes	no
LayerSkip	continued training	no / separate exit policy	partly / speculative	no	no
AttnRes pretraining	no	no	yes	no	yes
AR-Retrofit + ReSkip	**yes**	**no**	**yes**	**yes**	**yes**

closest conceptual antecedent, showing that recurrent CNNs can spend more steps on harder images to explain speed–accuracy behaviour. Recurrence also has a depth interpretation: finite recurrence can be unrolled into feedforward depth, but the recurrent form reuses a fixed substrate for flexible effective depth [44]. We take this as a design target, not a cognitive-faithfulness claim: cortical microcircuits and dendritic association [3, 6, 27] motivate intrinsic routing, while our implementation is a transformer retrofit.

In residual architectures, DenseNet [18] and Highway networks [43] modify residual flow; AttnRes [25] generalises residual combination with depth attention. To our knowledge, this is the first work to use AttnRes weights as a pre-execution skip signal and to retrofit an AttnRes-style routing path into pretrained transformers. In VLA, RT-2 [5], Octo [33], OpenVLA [23], and Pi0 [4] established the paradigm, while efficiency work has focused mainly on action chunking [50] and post-hoc compression. We contribute modality-aware adaptive depth for the VLM backbone.

8 Discussion

AR-Retrofit shows that a frozen pretrained transformer can acquire a pre-execution depth-routing path through a short identity-preserving fine-tune. ReSkip shows that this installed signal can be calibrated into input-dependent block execution, with ablation or drift checks determining which blocks are safe to skip. The current evidence is strongest for the retrofit mechanism and calibrated dynamic-depth behavior; broader backbone coverage, stronger end-to-end acceleration, and per-modality token-level calibration remain open.

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A Limitations and future work

Limitations. (i) Two retrofit targets from one VLM family; extension to InternVL / Gemma-VL / text-only LMs is planned. (ii) Runtime measurements are H100 / bf16 / batch 1 and use standard PyTorch inference paths; lower-end GPU / edge measurement is not yet done. (iii) The current RESKIP rule is block-level, inherited from ATTNRES; per-token routing and fused production kernels are future work. (iv) The VLA Pareto thresholds are sim-calibrated per scale, not per modality *within* a single rollout; per-modality, per-token-class calibration is the next sweep.

Future work: weight-shared AttnRes (RELOOP). A natural extension shares block weights across depth and differentiates each application via position-indexed pseudo-queries; the ATTNRES routing statistic can then serve both RESKIP's high-immediate-predecessor skip rule and a RELOOP halt rule based on low marginal change across recurrent applications. A 74M validation produced a smooth depth–quality Pareto (25% compute saved at $< 1\%$ ppl change); scaling and combining with the retrofit framework is open.

B Implementation details

B.1 Online softmax merge with skip

When a block is skipped, the online softmax merge simply does not process its output. The running state $(\mathbf{s}, m, e)$ is unchanged. The output is mathematically equivalent to an ATTNRES model trained without that block contributing—modulo the calibration of downstream $\mathbf{w}_l$.

B.2 Pseudo-query initialisation (Section 2.3 from-scratch)

For from-scratch training we use $\mathbf{w}_l \sim \mathcal{N}(0, 0.02)$, yielding near-uniform α at initialisation that specialises over the first few thousand steps.

B.3 Retrofit hyperparameters

AdamW with $\beta{=}(0.9, 0.95)$, weight decay 0, gradient clip 1.0, bfloat16. Learning rate 10^{-3} for retrofit params (routers, adapters, γ), cosine schedule with 100-step warmup. Sequence length 1536 (training), 2048 (evaluation). Loss weights $\lambda_{\text{kl}}{=}1$, entropy weight 0.02, KD temperature 1. The entropy term is subtracted from the loss, so it maximises routing entropy early in training rather than minimising it. Skip-branch sampling: one eligible block chosen uniformly at random per step. Adapter bottleneck rank $r{=}256$ (canonical); position-bias on router keys enabled; router temperature fixed at 1. γ-curriculum: $0 \to 1$ linearly over the first 30% of total steps (2B 5k/10k, 4B 5k), with ramp-fraction 0.5 on 4B $\times$10k to stabilise a late-stage $\gamma{=}1$ transition divergence (Appendix B.7). Total training wall-clock on a single H100: $\sim$22 min for 2B 5k, $\sim$44 min for 2B 10k, $\sim$54 min for 4B 10k.

B.4 Block granularity

For the final recipe we use $L{=}4$ layers per block on both scales: Qwen3-VL-2B has $N{=}7$ blocks and Qwen3-VL-4B has $N{=}9$ blocks. Earlier $L{=}2$ runs remain in the appendix as ablation controls, but $L{=}4$ is the canonical setting because it preserves the accuracy plateau while halving router calls relative to $L{=}2$. Each block receives one router, one residual adapter ($r{=}256$ default), and a scalar γ gate.

B.5 Identity-at-init details

With γ_n=0 and the adapter up-projection initialised at $\mathcal{N}(0, 0.02^2)$, the retrofit forward is arithmetically identical to the pretrained model (both produce $x_n = h_{n-1}$, which is passed into the unmodified block). We verified end-to-end: at n=500 on LAMBADA the retrofit and the base agree to within bfloat16 evaluation noise (acc 0.530 vs. 0.532; ppl 5.572 vs. 5.547); on MMMU and MMStar they are bit-equal at the answer-choice level. The smoke test at the token level shows max $|\Delta\text{logits}| = 0.375$ (bfloat16 SDPA noise) and 100% argmax agreement on our calibration prompts.

B.6 Skip under `use_cache=True`: correctness verification

We verify that the K/V-only skip path described in §2.2 produces an argmax-consistent output with the cacheless path. Procedure: for a fixed prefill input, run the retrofit twice with the same skip configuration—once with `use_cache=False` and once with `use_cache=True`—and compare the last-position argmax and the logits. We swept skip sets $\{[4], [4, 10], [4, 10, 12], [2, 4, 10]\}$ on H_r256_5k on Qwen3-VL-2B. In every configuration the last-position argmax matches between the two cache regimes, and the maximum logit delta is 0.19–0.37, identical to the bfloat16 SDPA jitter observed on the stock base with no retrofit and no skip. Multi-step autoregressive divergence starting around position 27–40 is an inherent property of HF + bf16 + SDPA that the stock base exhibits on the same prompts (measured on "Once upon a time" and "def fibonacci"); it is not a property of our skip path.

B.7 γ=1 transition stability

The γ curriculum is used to keep the frozen residual stream close to its pretrained operating point while the routed correction becomes active. Across the reported 2B and 4B canonical runs, we use a ramp that reaches γ=1 after at least 5,000 optimisation steps. This schedule was stable across the scales and block partitions reported in the paper and does not confound the quality comparisons in Appendix E.6.

C Retrofit ablations and inference machinery

C.1 Design controls

We compared the γ-gated residual injection against three simpler designs: observer-only routing, block-input interpolation, and direct ATTNRES replacement with informed pseudo-query initialisation. These controls either made the routing weights non-load-bearing, moved the frozen backbone too abruptly away from its pretrained residual stream, or required substantially more tuning than the short retrofit budget. The γ-gated injection of §2.2 is the stable recipe used in all main experiments.

C.2 Adapter-rank ablation

Wall-clock latency is essentially unchanged across ranks (±2% at all measured sequence lengths): the adapter's 2048→r→2048 bottleneck is a small contributor compared to the router stack/softmax/einsum cost. Rank therefore affects quality but not speed. The canonical r=256 was selected from the γ→1 ablation of Appendix C.3; at equal step budgets lower ranks leave more of the MMBench drop on the table without compensating latency savings.

C.3 γ→1 retrofit ablation

We swept variants of the canonical γ→1 recipe along four axes: adapter rank $r \in \{32, 64, 128, 256\}$, training steps $\in \{5\text{k}, 10\text{k}, 20\text{k}\}$, LLaVA-Instruct fraction of the mix $\in \{50\%, 80\%\}$, and γ ramp-fraction $\in \{0.3, 0.7\}$. All other hyperparameters (loss, optimizer, schedule family, seed) are held constant.

Over-training signature at γ=1. Holding (r=256, 50% VLM, fast ramp) fixed and varying only the total step count, MMBench is strictly monotonic in steps: 5k → 0.710 (preserved), 10k → 0.587 (−14pp), 20k → 0.513 (−21pp). LAMBADA peaks at 10k (0.586) and regresses at 20k (0.568), so longer training buys no text-domain gain either. The γ=1 regime has a narrow compute sweet spot; past it the router+adapter over-specialise to the training distribution and the frozen backbone, unable to adapt further, loses its multimodal calibration.

Table 10: **Retrofit ablation** on Qwen3-VL-2B. All rows use the γ-curriculum $0 \to 1$ over the first 30% of steps unless noted. "n=300 MMBench noise floor ± 1 question" corresponds to ± 0.33pp.

Name	γ sched	rank	steps	VLM%	ramp	LAMBADA	HellaSwag	MMBench
Base Qwen3-VL-2B	—	—	—	—	—	0.532	0.506	0.727
Canonical	$0 \to 1$ fast	$\mathbf{256}$	$\mathbf{5}$**k**	$\mathbf{50}$	$\mathbf{0.3}$	$\mathbf{0.576}$	$\mathbf{0.522}$	$\mathbf{0.710}$
+ low rank	$0 \to 1$ fast	64	10k	50	0.3	0.570	0.530	0.697
+ very low rank	$0 \to 1$ fast	32	10k	50	0.3	0.568	0.526	0.683
+ low rank, VLM-heavy	$0 \to 1$ fast	64	10k	80	0.3	0.560	0.524	0.660
+ mid rank, VLM-heavy	$0 \to 1$ fast	128	10k	80	0.3	0.564	0.520	0.653
+ VLM-heavy	$0 \to 1$ fast	256	10k	80	0.3	0.560	0.520	0.617
+ longer, 10k	$0 \to 1$ fast	256	10k	50	0.3	0.586	—	0.587
+ slow ramp	$0 \to 1$ slow	256	10k	50	0.7	0.568	0.520	0.517
+ longer, 20k	$0 \to 1$ fast	256	20k	50	0.3	0.568	0.520	0.513

Rank and data are subordinate. Halving rank at 10k recovers some MMBench (r=32: 0.683, r=64: 0.697), but never reaches the 5k-r=256 level (0.710). Pushing the mix to 80% LLaVA *worsens* MMBench at every rank, indicating the regression is router over-adaptation to *any* narrow training distribution, not insufficient visual exposure.

Interpretation. These ablations are used only to set the canonical recipe. They are not the main evidence for ReSkip; the load-bearing evidence is that the installed routing path participates in the forward computation and supports the calibrated skip rule in §4.2.

C.4 Per-block skip importance

Running single-block removals on the canonical 2B retrofit gives the eligibility set used in the main text, $\mathcal{P}=\{1, 4\}$ (Tab. 3). The same qualitative ranking pattern appears across the 340M from-scratch and 2B retrofit settings: some blocks near the embedding or late residual-refinement stages are costly to remove, while selected mid-depth positions are safer candidates after per-model calibration.

C.5 Calibration set (dynamic skip)

Calibration uses 32 held-out LAMBADA prefixes (truncated to 512 tokens each). Per-block $w_{\text{recent}}(n)$ samples are collected with a single retrofit forward and the empirical quantile at q is used as τ_n. We verified that varying the calibration draw shifts τ_n by < 0.01 in absolute terms at the chosen q=0.95. For VLA deployment, the identical calibration pipeline applied to the VLA in-backbone forward (`retrofit/eval/calibrate_vla_thresholds.py`) produces thresholds byte-identical to the LAMBADA-calibrated set on matched inputs, confirming that the VLA and VLM forwards instantiate the same router.

C.6 Canonical data mixture

Table 11 gives the final SFT mixture used for the canonical retrofit. The mix is intentionally VLM-anchored, with a smaller math/reasoning component to strengthen deliberate reasoning without moving the frozen VLM backbone away from visual instruction following.

This mixture is the only data recipe used for the canonical results in the main paper.

C.7 Two-phase forward with dynamic skip (algorithm)

Table 11: **Canonical retrofit SFT mixture.** The same source proportions are used for the 2B and 4B canonical L=4 recipes in Tab. 2.

Source	Proportion	Role
LLaVA-OneVision-Data [28]	60%	visual instruction anchor
UltraChat [11]	20%	general instruction following
NuminaMath-CoT [1]	10%	mathematical reasoning traces
OpenThoughts-114k [17]	10%	open-ended reasoning traces

Algorithm 1 Two-phase AttnRes forward with dynamic ReSkip.

Input: embeddings $\mathbf{h}_0$; block functions $f_1, \ldots, f_N$; pseudo-queries $\{\mathbf{w}_n\}$; eligible $\mathcal{P}$; thresholds $\{\tau_n\}$; max skips $M_{\max}$.
Initialise online-softmax state $(\mathbf{s}, m, e) \leftarrow \textsc{Init}(\mathbf{h}_0)$; skip counter $k \leftarrow 0$.
for $n = 1$ to N **do**
  Compute $\alpha_{\cdot\to n}$ over completed block outputs (*phase 1*).
  Form routed-correction input $\mathbf{x}_n$ as in Eq. 2.
  $w_{\text{recent}}(n) \leftarrow \mathbb{E}_{\text{token}}[\alpha_{n-1\to n}]$.
  if $n \in \mathcal{P}$ **and** $k < M_{\max}$ **and** $w_{\text{recent}}(n) > \tau_n$ **then**
    Skip: set $\mathbf{h}_n \leftarrow \mathbf{x}_n$; under `use_cache=True`, run K/V-only slice on each constituent layer; $k \leftarrow k + 1$.
  else
    Execute block n on the merged input; update $(\mathbf{s}, m, e)$ with $\mathbf{h}_n$.
  end if
end for
Output: final hidden state $\mathbf{s}/e$.

D Phase 1: 340M from-scratch validation

D.1 Motivation: AttnRes cost from-scratch is prohibitive

We measured forward-pass latency on 1×H100 / bf16 / batch 1 for two 340M models trained under identical FineWeb-Edu 100BT recipes: a vanilla standard-residual transformer and an AttnRes transformer with N=8 blocks. Block-AttnRes adds $+78\%$ (8192 seq) to $+128\%$ (1024 seq) wall-clock vs. vanilla. At the 2B VL scale a stock Qwen3-VL-2B forward already takes $\sim$25.6 ms at seq 256 on 1×H100, so a 35–45% block-level AttnRes overhead would push past common 50 Hz / 20 Hz robotic control budgets. Pretraining a 2B AttnRes-VLM costs at least what the matched standard-residual base costs ($\geq$ 10k–25k H100-h reading published Qwen3-VL / InternVL3 / OpenVLA / LLaVA-OneVision tech reports), so the only practical access path is a retrofit on the already-trained standard model.

D.2 Phase 1: cross-scale validation

The 340M from-scratch AttnRes (§2.3, Table 1) is the load-bearing existence proof. A 110M cross-scale check on the same FineWeb-Edu recipe with an 8-block partition reproduces the zero-degradation pattern at a safer operating point (q=0.97, $M_{\max}$=1 vs. 340M's q=0.85, $M_{\max}$=2); skip trigger rate is lower at 110M, consistent with larger models carrying more redundant computation. 1.3B / 2B from-scratch are not load-bearing because the retrofit (§4) directly targets a pretrained 2.13B model, answering the 2B-scale question without paying the pretrain bill.

Table 12: RESKIP versus representative adaptive-computation methods.

Method	Auxiliary network	Routing is	Train-time changes	Architectural changes
CALM [40]	exit classifiers	per-token	yes	small
Mixture-of-Depths [39]	router	per-token	yes, capacity loss	yes
Static pruning [16]	none	static	no	block removed
LayerSkip [13]	spec. decoder	per-token	yes, cont. pretrain	decoding path
RESKIP (ours)	**none**	**per-batch**	**none**	**none**

Table 13: Dynamic skip as a function of $\mathcal{P}$ on 340M from-scratch ATTNRES ($M_{\max}$=2, q=0.85). Picking by I alone ($\{5\}$) or A alone ($\{3\}$) leaves speed or quality on the table; combining the two ($\{3, 5\}$) is strictly best.

$\mathcal{P}$	PPL ratio	Speedup
$\{5\}$ (I-only)	0.993	1.14×
$\{3\}$ (A-only)	1.009	1.02×
$\{2, 3\}$	1.009	1.17×
$\{4, 5\}$	1.008	0.97×
$\{3, 4, 5\}$	0.993	1.02×
$\{2, 3, 4, 5, 6\}$	1.40	1.16×
$\{3, 5\}$ **(ours)**	**0.991**	**1.19**×

D.3 RESKIP method comparison and full Pareto / latency at 340M

D.4 RESKIP position-set ablation at 340M

D.5 Decision rule: dynamic vs. static-rate-matched vs. random

The position-set ablation above answers *which* blocks to skip; this subsection answers *whether* the input-dependent decision matters at all. We hold the 340M from-scratch ATTNRES weights fixed, fix $\mathcal{P}$=$\{3, 5\}$ and $M_{\max}$=1 (block-level skip rate $\leq 12.5\%$), and only vary the runtime decision rule:

- **B0.** No-skip upper bound (full ATTNRES forward).
- **B1.b/B2.a.** Dynamic, fire when $w_{\text{recent},n} > \tau_n$ (our RESKIP signal at L=1 block granularity).
- **B1.c, B1.d.** Static, every-token skip at P=3 or P=5 (rate-matched, 12.5%).
- **B1.e/B2.d.** Random, per-call uniform draw from $\{$keep-P=3, keep-P=5$\}$ (rate-matched, 12.5%).
- **B2.b.** Dynamic, fire when block-1 entropy $H_n < \tau_n$ (router-confidence rule).
- **B2.c.** Dynamic, fire when $w_{\text{recent},n} - w_{\text{embed},n} > \tau_n$ (relative-recent rule).

All three thresholds are calibrated as the q=0.5 per-position quantile on 32×8192 FineWeb-Edu tokens (§C.5). Results in Tab. 14; observed skip rates on the eval distribution in Tab. 15.

Conclusion. At a fair rate-matched comparison ($\sim$12% fired skips), *input-dependent* dynamic skip (B2.c) preserves LAMBADA at 0.345 while every static or random alternative degrades to 0.22–0.26 (-8 to -13pp). The same ordering holds on HellaSwag, PIQA, ARC-easy and OpenBookQA. The MoD-style "you might be getting a free lunch from any same-rate schedule" critique is rejected: at 340M from-scratch, schedule choice matters and the ATTNRES routing weights carry the load.

Threshold-transfer caveat. Two of the three dynamic rules (B1.b `recent_weight_gt` and B2.b `entropy_lt`) fire far below the 12.5% FineWeb-Edu calibration target on the LAMBADA distribution (Tab. 15). Their near-no-skip accuracy is therefore not a positive datapoint for those specific rules; it is consistent with "the threshold protected the

Table 14: **Decision-rule and threshold-rule ablation on** 340**M from-scratch** **ATTNRES**, lm-evaluation-harness with $\mathcal{P}$=$\{3, 5\}$, $M_{\max}$=1. Dynamic rules whose threshold actually fires on the eval distribution (B2.c) preserve LAMBADA at 0.345 vs. static / random at the same $\sim$$12\%$ rate (0.22–0.26). Dynamic rules whose calibrated thresholds rarely cross on lm-eval data (B1.b at 3.1% observed; B2.b at 0.0%, see Tab. 15) are nearly equivalent to no-skip and are not the rate-matched comparison.

Cell	LAMBADA	HellaSwag	PIQA	ARC-e	OpenBookQA
B0. no-skip (upper bound)	0.4054	0.4607	0.6893	0.5438	0.3580
Dynamic, calibrated q=0.5 *on FineWeb-Edu*					
B1.b. recent_weight_gt	0.4011	0.4534	0.6839	0.5412	0.3580
B2.b. entropy_lt	0.4036	0.4608	0.6839	0.5417	0.3580
B2.c. recent_minus_embed_gt	**0.3445**	**0.4471**	**0.6774**	**0.5391**	**0.3540**
Static / random, rate-matched at 12.5%					
B1.c. static skip P=3 every	0.2624	0.3968	0.6610	0.5206	0.3300
B1.d. static skip P=5 every	0.2189	0.4223	0.6638	0.5253	0.3260
B1.e/B2.d. random $\{P$=3 OR P=$5\}$	0.2416	0.4028	0.6420	0.5101	0.3300

Table 15: **Observed skip rate on** 8×512 **LAMBADA tokens.** The calibration set (FineWeb-Edu) and the eval set (LAMBADA) have different routing-statistic distributions; B2.b's `entropy_lt` threshold never fires on LAMBADA and B1.b fires only 3.1%, so their headline-table accuracy is misleadingly close to no-skip. B2.c is the strategy whose calibration target ($\sim$$12.5\%$) actually transfers, and is therefore the load-bearing rate-matched comparison against B1.c/d/e in Tab. 14.

Cell	Calibration target	Observed (LAMBADA)	Notes
B0	0%	0.00%	sanity check, τ_n=∞
B1.b	$\leq 12.5\%$	3.12%	τ_3=0.4645, τ_5=0.4010
B2.b	$\leq 12.5\%$	0.00%	τ_3=0.8631, τ_5=0.8764
B2.c	$\leq 12.5\%$	$\mathbf{10.94\%}$	τ_3=0.1655, τ_5=0.2472
B1.c	12.5%	12.50%	forced, every-token P=3
B1.d	12.5%	12.50%	forced, every-token P=5
B1.e	12.5%	per-batch toggle	uniform $\{P$=3, P=$5\}$

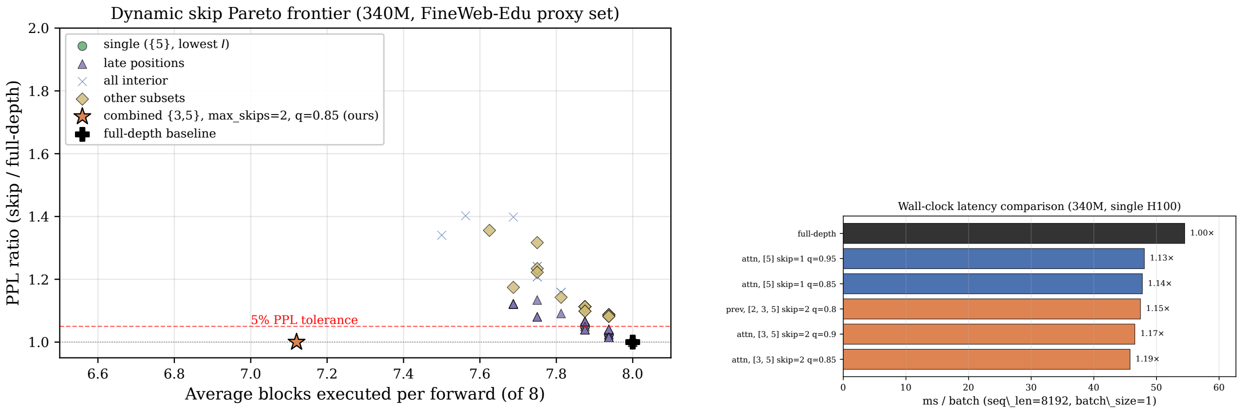
output by accidentally not firing." The load-bearing rate-matched comparison is B2.c `recent_minus_embed_gt` (which does fire close to target) versus the static/random alternatives. We treat the two non-firing rows as a sensitivity result: RESKIP's safety margin under threshold/distribution mismatch is high (no degradation when the rule rarely fires), but deployment requires per-distribution threshold calibration, as already noted for the cross-modality VLA case (§F.3).

D.6 RESKIP full Pareto and latency curves at 340M

E Retrofit Pareto and breakdowns

E.1 LoRA baselines

These auxiliary controls ask whether adding a similar number of trainable parameters through a standard adaptation path is enough to reproduce the text gains. They are pilot controls on an earlier $50/50$ UltraChat + LLaVA mix, not a final-mixture baseline for Tab. 2. LoRA r=32 on q, v: LAMBADA acc 0.540 / 0.516 (two seeds), HellaSwag 0.510 / 0.524. LoRA r=16 on q, k, v, o: LAMBADA 0.534, HellaSwag 0.492. LoRA r=8 on MLP: LAMBADA 0.514, HellaSwag 0.510. Mean LAMBADA across the four runs is 0.526, -0.6pp below base (0.532) and -5.0pp below the matched $\gamma \to 1$ retrofit on that mix (0.576). Mean HellaSwag 0.509 is $+0.3$pp over base versus retrofit's $+1.6$pp. These controls rule out a simple parameter-count-only explanation under a related SFT setting, but they do not prove that all



(a) Accuracy–compute Pareto. $\mathcal{P}=\{3,5\}$ (orange star) sits below the 5% PPL tolerance.

(b) Wall-clock at seq 8192. $\{3,5\}/M=2/q=0.85$ reaches $1.19\times$.

Figure 3: ResKip on 340M FineWeb-Edu: position-calibrated dynamic skip with $\mathcal{P}=\{3,5\}$ strictly dominates the single-position baseline at zero benchmark drop.

quality gains come uniquely from routing.

E.2 Static-pruning baseline at the VLM scale (Gromov drop-k)

The 340M decision-rule ablation (Tab. 14) shows that *single-position* static skip and random skip both degrade on language tasks. The complementary question at the VLM scale is whether *full-layer* static pruning—the standard "remove the least useful blocks" baseline used by Gromov et al. [16]—can match the accuracy floor that the retrofit operates above. We rank the 28 Qwen3-VL-2B layers by the Gromov angular-distance criterion (computed on a FineWeb-Edu calibration set with the LM-head only; no fine-tuning), remove the k lowest-distance layers, and re-evaluate on `lmms-eval`. Tab. 16 reports drop-4 (layers $\{23, 24, 25, 26\}$) and drop-8 (additionally $\{12, 13, 14, 22\}$).

Table 16: **Static layer pruning on Qwen3-VL-2B** (Gromov-style angular-distance ranking, no retraining), `lmms-eval` full splits. Companion to the 340M in-block decision-rule ablation in Tab. 14: spans the *full-layer* vs. *single-position-within-block* skip dimension. Pruning 4 of 28 layers (14%) costs −16pp AI2D, −6pp MMMU, −5pp MMStar, −65pp OCRBench; pruning 8 (29%) sharply degrades every cell. The retrofit (Tab. 2) operates at 0 removed layers, 0pp loss, and additionally lifts the base; ResKip on the retrofit at q=0.85 holds within 1pp on LAMBADA (Tab. 17).

Configuration	AI2D	MMMU	MMStar	OCRBench	RWQA
Qwen3-VL-2B (base, L_{rm}=0)	0.736	0.414	0.536	0.772	0.648
Gromov drop-4 (layers $\{23, 24, 25, 26\}$)	0.5732	0.3556	0.4906	0.1240	0.3791
Gromov drop-8 (drop-4 $\cup$ $\{12, 13, 14, 22\}$)	0.0683	0.2389	0.0190	0.0030	0.1451
+ AR-Retrofit v3 (L=4, 10k; Tab. 2)	**0.758**	**0.432**	**0.536**	**0.814**	**0.661**

The pattern matches the in-block 340M observation: any *unconditional* block removal (single-position static at 340M, full-layer Gromov at 2B) destroys benchmark accuracy at the VLM scale even at modest pruning fractions, while input-dependent skip on top of an AttnRes routing structure preserves it. We do not attempt to retrain the pruned model; the intent is a free static-baseline floor for the cost-quality trade-off, not a tuned competitor.*

*Pruned cells additionally exhibit POPE generation drift below the strict yes/no surface (drop-4 pope-acc 0.021, drop-8 0.500 via accidental all-"yes" decoding), so we drop POPE from the table.

E.3 RESKIP on the canonical retrofit: text Pareto and cross-modality consistency

On the canonical $L=4$ v3 10k retrofit, applied to LAMBADA-500 as the language-modality cross-check of the LIBERO Pareto (Tab. 6), RESKIP produces the same shape as on action: lossless at the conservative end, collapsing once q moves below the action distribution's mean. Same $\mathcal{P}=\{1, 4\}$, $M_{\max}=2$ as the 2B VLA cell.

Table 17: **Cross-modality RESKIP consistency.** LAMBADA-500 on the canonical $L=4$ v3 retrofit, $\mathcal{P}=\{1, 4\}$, $M=2$, τ calibrated on 32 held-out LAMBADA prefixes at quantile q. Same operating-point structure as the LIBERO 4-suite Pareto: near-lossless above the calibration distribution mean ($q \geq 0.85$); sharp degradation below, where the threshold instructs the router to skip blocks it is normally executing.

q	LAMBADA acc	LAMBADA ppl	Δ acc vs no-skip	avg. skips / forward
no-skip ($M=0$)	0.5700	4.526	—	0.00 / ≤ 7
0.85	0.5600	5.258	-1.0	0.19 / 2
0.50	0.4120	12.550	-15.8	1.06 / 2
0.30	0.3900	14.005	-18.0	1.17 / 2

E.4 Wall-clock latency: full table including earlier $L=2$ and VLA in-backbone

Tab. 4 in the main body reports the canonical $L=4$ result. Tab. 18 below shows the full set: the earlier $L=2$ partition (14 blocks at 2B) carries a $1.39\times$ structural cost over stock Qwen3-VL-2B because each token pays 14 router calls, the VLA in-backbone forward (the same retrofit run inside the OFT trainer's backbone forward, §5) tracks the VLM retrofit within 1% at both partitions, and `torch.compile` closes the $L=2$ residual to $1.16\times$ but is more dramatic on $L=4$ where the router count is halved. This table is the basis for the canonical-partition switch from $L=2$ to $L=4$.

Table 18: **Forward-pass latency, full sweep** on $1\times$H100, bf16, batch 1, median of 20 runs after 5 warmup. Eager rows: stock-HF base vs. retrofit at the indicated L. Compile rows wrap both with `torch.compile`, `dynamic=False`; `reduce-overhead` captures CUDA graphs, `max-autotune` additionally tunes per-kernel matmuls. The VLA in-backbone row is the canonical $L=4$ retrofit loaded into the OFT trainer's backbone forward (a separate code path) — within 1% of the VLM retrofit cell, confirming the per-block router stack, not the VLA harness, is the structural cost. *Bold:* the canonical operating point of the paper.

Configuration	seq	mode	base (ms)	retrofit (ms)	retrofit / base
(1) $L=2$ retrofit (earlier)	1024	eager	15.42	21.25	$1.378\times$
(2) $L=2$ retrofit (earlier)	2048	eager	26.03	36.14	$1.388\times$
(3) $L=4$ retrofit (canonical)	1024	eager	14.91	16.66	$1.117\times$
(4) $L=4$ retrofit (canonical)	2048	eager	25.49	28.53	$1.119\times$
(5) $L=2$ retrofit	1024	`reduce-overhead`	9.31	10.78	$1.158\times$
(6) $L=2$ retrofit	2048	`reduce-overhead`	21.25	24.45	$1.151\times$
(7) $L=4$ retrofit	2048	`reduce-overhead`	21.31	22.50	$1.056\times$
(8) $L=4$ retrofit	**2048**	**`max-autotune`**	**21.81**	**22.43**	$\mathbf{1.029}\times$
(9) VLA in-backbone $L=4$ retrofit	2048	eager	25.49	28.79	$1.130\times$

The structural per-block router stack (`stack`$\to$`RMSNorm`$\to$`softmax`$\to$`einsum` over N completed blocks) is small but visible in eager inference. Adapter rank ablation (Appendix C.2) confirms the adapter is not the bottleneck ($\leq 2\%$). `torch.compile` (rows 5–8) collapses the small router gemms into a captured graph and tunes the matmul kernels for retrofit's small shapes, removing most of the $L=4$ structural gap. The remaining 2.9% is the cost of the installed routing path at the canonical operating point; fusing the router and skip decision into a single production kernel is an implementation direction rather than a change to the method.

E.5 Compile vs. eager: accuracy parity

The compiled-overhead result of §4 requires that `torch.compile` preserve the retrofit's accuracy. We verified two parity tests on the canonical L=4 v3 10k retrofit.

LAMBADA-500 **accuracy parity.** Running LAMBADA-500 with the retrofit wrapped in `torch.compile` using default mode and dynamic shapes: acc 0.5720 compiled vs. 0.5700 eager (Δ=+0.20pp), ppl 4.534 vs. 4.526. Per-target argmax agreement against eager: 98.60% — the residual 1.4% are tokens where eager and compiled both fall within bf16 SDPA jitter and the rank-1/rank-2 logits flip.

Per-token logit parity. On real prompt tokens with `mode="reduce-overhead"` (the inference-time mode used for the speed table): per-token argmax agreement 98.51%, max $|\Delta\text{logits}| = 0.50$, RMSE 0.060 — same magnitude as the cache-on/cache-off SDPA jitter on the stock base (Appendix B.6). Compile preserves the retrofit's forward to within bf16 evaluation noise; the 1.029× `base_compiled` number is at matched accuracy.

E.6 Block partition sweep

Table 19: Block partition sweep (v3 recipe, r=256, γ-curriculum). L is layers-per-block, N=L_{total}/L. Per-layer L=1 underperforms by 2–9pp; the plateau $L \in \{2, 4, 7\}$ on 2B reproduces Kimi Team et al. [25]'s $S \in \{2, 4, 8\}$. We recommend L=4 on both scales because it preserves quality while reducing router calls.

Scale	L	N	LAMBADA	ΔL	HellaSwag	MMBench	MMMU	MMStar	AI2D	OCRBench	RWQA
2B base	—	—	0.532	—	0.506	75.77	0.414	0.536	0.736	0.772	0.648
2B retrofit	1	28	0.5645	+3.3	0.490	76.20	0.388	0.499	0.743	0.809	0.652
2B retrofit	2	14	0.5755	+4.4	0.494	77.23	0.426	0.532	0.748	0.803	0.663
2B (rec.)	**4**	**7**	**0.5650**	**+3.3**	**0.500**	**78.87**	**0.432**	**0.536**	**0.758**	**0.814**	0.661
2B retrofit	7	4	0.5155	−1.7	0.492	77.49	0.427	0.530	0.756	0.808	0.656
4B base	—	—	0.576	—	0.562	83.33	0.490	0.624	0.819	0.819	0.715
4B retrofit	1	36	0.5575	−1.9	0.523	83.33	0.497	0.538	0.783	0.768	0.694
4B retrofit	2	18	–	–	–	84.28	0.523	0.587	0.816	0.813	0.708
4B (rec.)	**4**	**9**	**0.6625**	**+8.7**	**0.552**	**85.22**	0.521	**0.632**	**0.825**	**0.824**	**0.718**
4B retrofit	6	6	0.6540	+7.8	0.554	84.79	**0.531**	0.623	0.817	0.824	0.715

E.7 MMStar subcategory breakdown

Table 20: MMStar subcategory breakdown for the v3 retrofit. The math, logical, and sci&tech triple is the "reasoning over images" cell where the retrofit shows the largest gain (+7.9pp math on 2B; +3.9pp on 4B). Perception cells (coarse / fine / instance) are at parity or a small regression, consistent with a router that lifts deliberate-reasoning paths rather than altering early perception.

Configuration	math	logical	sci&tech	coarse	fine	instance
Qwen3-VL-2B (base)	0.413	0.429	0.408	0.714	0.520	0.710
+ AR-RETROFIT v3 (2B, L=4)	**0.492**	**0.432**	0.353	0.734	0.505	0.683
Δ vs. 2B base	**+7.9**	+0.3	−5.5	+2.0	−1.5	−2.7
Qwen3-VL-4B (base)	0.549	0.626	0.465	0.788	0.611	0.705
+ AR-RETROFIT v3 (4B, L=4)	**0.588**	0.602	**0.467**	0.812	0.606	0.714
Δ vs. 4B base	**+3.9**	−2.4	+0.2	+2.4	−0.5	+0.9

F VLA appendix

F.1 VLA seed variance on 2B

We ran two seeds for the two 2B suites where Path 0 / Path B are within evaluation noise on a single seed; `libero_spatial` and `libero_object` are single runs. Aggregating both repeated rollouts gives Path 0 96.05 and Path B 96.75 ($\Delta = +0.70$pp), while the Table 5 operating-point values give $\Delta = +1.30$pp. Both summaries preserve the ranking Path B $>$ Path 0. The main text therefore treats LIBERO as supporting transfer evidence rather than as the primary contribution.

Table 21: Per-seed Qwen3-VL-2B numbers. Seed 1 is the initial run; seed 2 is a fresh environment-reseeded rollout of the same trained checkpoint. The last column shows the values used in Table 5.

Suite	Policy	Seed 1	Seed 2	Used in Tab. 5
`libero_goal`	Path 0 (30k)	97.6	97.4	97.4
`libero_goal`	Path B (30k)	97.4	98.6	98.6
`libero_10`	Path 0 (30k)	92.8	91.4	91.4
`libero_10`	Path B (30k)	92.6	90.6	92.6

F.2 VLA landscape: open VLAs on standard benchmarks

Table 22: Open VLA landscape on standard benchmarks. LIBERO entries are success rates (%) per task suite. Following common practice, π_0 and Octo-Base LIBERO numbers are taken from OpenVLA-OFT [24], which re-ran all three policies under a uniform LIBERO protocol. Our rows use the 30k Path B numbers from Table 5.

Model	#Params	Spatial	Object	Goal	Long-10	ref.
Octo-Base	93M	78.9	85.7	84.6	51.1	[33, 24]
OpenVLA	7B	84.7	88.4	79.2	53.7	[23, 24]
TraceVLA	7B	84.6	85.2	75.1	54.1	[51]
SpatialVLA	4B	88.2	89.9	78.6	55.5	[37]
π_0	3B	**96.8**	**98.8**	95.8	85.2	[4, 24]
Qwen3-VL-2B + AR-Retrofit + OFT (*ours*)	2B	**97.8**	99.6	**98.6**	92.6	this work
Qwen3-VL-4B + AR-Retrofit + OFT (*ours*)	4B	94.6	**99.8**	**98.2**	**94.2**	this work

F.3 VLA ReSkip setup: per-block drift and eligible-set selection

The ReSkip eligible set $\mathcal{P}$ for each scale is selected from a per-block ablation *on the trained Path B policy*, not on the VLM retrofit alone, because the VLA action stream activates a different routing distribution than the language stream. We measure per-block action-drift MSE on a 24-sample sim trajectory: the policy is run with no skip and with each block individually skipped; the per-step action MSE between the skipped and no-skip rollouts is the per-block "cost of removing this block on the action distribution". The two safest blocks per scale form $\mathcal{P}$.

2B Path B 30k per-block action-drift MSE (sorted ascending, in units $\times 10^{-3}$): block 1: 0.4; block 4: 34.0; block 0: 58.1; block 5: 87.0; remaining blocks $>$100. Selecting the two lowest: $\mathcal{P}_{2B} = \{1, 4\}$.

4B Path B 30k per-block action-drift MSE: block 1: 0.6; block 2: 11.0; block 0: 34.0; block 3: 63.0; remaining $>$120. Selecting: $\mathcal{P}_{4B} = \{1, 2\}$.

Per-scale eligibility. The eligible set is selected separately for each model scale because the action-stream routing distribution and per-block drift are scale dependent. This is the per-scale half of the modality-aware skip protocol: α provides the online signal, while action-drift calibration selects the safe candidate blocks for the policy being deployed.

Threshold calibration. Per-block thresholds τ_n are the q-th empirical quantile of $w_{\text{recent},n}$ over $31{,}286$ (2B) / $31{,}257$ (4B) sim-rollout records (one record per decoded action token across many trajectories). The Tab. 6 numbers use this protocol. A separate "Method A" that calibrates on retrofit pretokenized data (as if the VLA were just a long-context LM) lands at a conservative operating point that mimics $q{=}0.99$ on the action distribution; we use it as a sensitivity comparator in our ablations and recommend the action-distribution-calibrated thresholds for any deployment.

Cross-modality threshold calibration. Language-calibrated thresholds and action-calibrated thresholds correspond to different effective skip rates because the action stream has a different w_{recent} distribution. We therefore calibrate τ_n on the action distribution for LIBERO rather than transferring the LAMBADA thresholds unchanged. This is the same calibration principle used in the main method: routing decides when to skip, while the eligible set and thresholds are selected on the deployment distribution.

F.4 VLA planned follow-ups

Per-modality, per-token-class RESKIP. The Tab. 6 thresholds are calibrated on the action distribution as a whole; an obvious refinement is per-modality $(\mathcal{P}^{(m)}, \tau_n^{(m)}, M_{\max}^{(m)})$ within a single rollout (vision tokens, language tokens, action tokens). Working hypothesis: vision α concentrates on early blocks, while action α spreads to late blocks and benefits from more conservative late-block skipping. *Fused routing kernels.* A static-graph variant of the skip rule could further reduce the installed-path runtime. *Edge hardware.* Wall-clock is H100 / bf16; RTX 4090 / Jetson Orin direct measurement is planned.

G NeurIPS Paper Checklist

1. **Claims.** Yes. The abstract, introduction, and conclusions state the main claims and separate the retrofit contribution from calibrated skipping, VLA transfer, and systems characterization.
2. **Limitations.** Yes. Appendix A lists architecture coverage, hardware scope, block-level routing granularity, and per-modality calibration limits.
3. **Theory assumptions.** Not applicable. The paper is empirical and algorithmic; all equations define implemented objectives or routing rules.
4. **Experimental reproducibility.** Yes. Sections 2.3–5.1 and the appendix specify model scales, block partitions, training steps, datasets, calibration rules, and evaluation protocols.
5. **Open access to code and data.** Yes. Experiments use public datasets and benchmarks; implementation details and release plan are described under the anonymous submission policy.
6. **Compute.** Yes. Main text and appendices report GPU type, training steps, rough GPU-minutes / GPU-hours, and latency measurement settings.
7. **Dataset and benchmark provenance.** Yes. All public datasets and benchmarks are cited; LIBERO and lmms-eval protocols are described with rollout counts or split usage.
8. **Human subjects.** Not applicable. No new human-subject data are collected.
9. **Privacy.** Not applicable beyond the privacy considerations of the cited public datasets.
10. **Licenses.** Yes. Public datasets/models are cited; release artifacts will include license metadata.
11. **Broader impacts.** Yes. The work targets lower inference cost and adaptive computation in pretrained models; no safety-critical deployment is claimed.
12. **Safeguards.** Yes. VLA thresholds are calibrated per model and action distribution; cross-modality transfer failures are explicitly documented as a deployment risk.